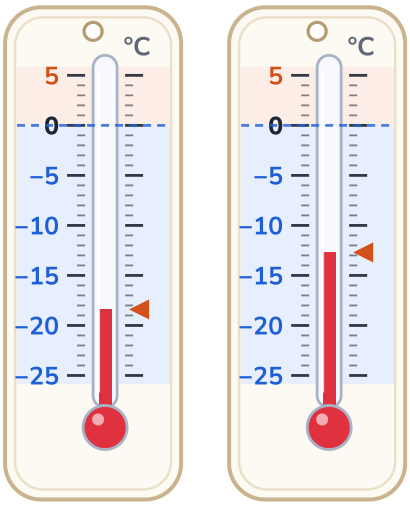
Haqiqiy sonlar ustida amallar

1-QISM

Qiziqtiramiz

1-SAVOL · MUZLATGICH

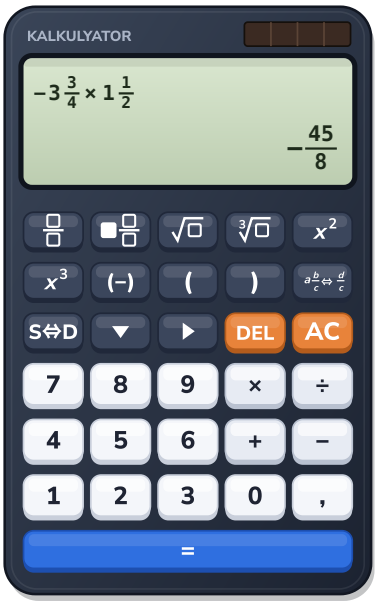
Muzlatgich kamerasida harorat −18,4 °C edi. Eshigi ochiq qolib, harorat 5,7 °C ga ko'tarildi. Termometr endi nechani ko'rsatadi?



Oldin Keyin

2-SAVOL · KALKULYATOR

Nodira kalkulyatorda $\left(-3\frac{3}{4}\right) \cdot 1\frac{1}{2}$ ni hisobladi. Ekranda kasr chiqdi. Javobni o'nli kasr ko'rinishida ko'rish uchun qaysi tugmani bosadi?





Manfiy sonlarni qo'shish va ko'paytirishni bilasiz. Kasr va o'nli kasr bo'lsa, qoidalar o'zgaradimi?

ESLAB OLAYLIK

- Ayirish — qarama-qarshi sonni qo'shish: $4 - (-9) = 13$. Ko'paytirish va bo'lishda ishoralar bir xil bo'lsa — musbat, har xil bo'lsa — manfiy: $(-6) \cdot (-2) = 12$, $15 : (-3) = -5$.
- Umumiy maxraj — maxrajlarning EKUKi: $\frac{1}{6} + \frac{3}{4} = \frac{2}{12} + \frac{9}{12} = \frac{11}{12}$.
- $\frac{2}{7}$ ga teskari son — $\frac{7}{2}$, chunki $\frac{2}{7} \cdot \frac{7}{2} = 1$.

2-QISM

Sinab ko'ramiz

1-TOPSHIRIQ · BUTUN SONLAR VA KASRLAR

Har juftni hisoblang va solishtiring:

a) $-5 + 2$ va $-\frac{5}{7} + \frac{2}{7}$ b) $3 - 8$ va $\frac{3}{10} - \frac{8}{10}$ c) $-4 - (-1)$ va $-0{,}4 - (-0{,}1)$

Nima o'xshash?

2-TOPSHIRIQ · VERGUL QAYERGA QO'YILADI?

Oddiy kasrga aylantirib hisoblang: a) $0{,}3 \cdot 0{,}2$ b) $1{,}2 \cdot 0{,}4$ c) $0{,}05 \cdot 0{,}7$

Ko'paytuvchilarda verguldan keyin nechta raqam bor edi? Natijada-chi?

3-TOPSHIRIQ · BIR XIL BO'LINMALAR

Hisoblang: $48 : 6$, $480 : 60$, $4{,}8 : 0{,}6$. Uchala bo'linma tengmi? Nega?

3-QISM

Tushunamiz

Endi amallarni $\frac{a}{b}$ ko'rinishidagi ratsional sonlar ustida bajaramiz (a va b — butun sonlar, $b \neq 0$): manfiy kasrlar, manfiy aralash sonlar va manfiy o'nli kasrlar ham shunga kiradi.

TA'RIF · QISQARMAS KASR

Agar oddiy kasrning surati va maxrajining EKUBi 1 ga teng bo'lsa, kasr qisqarmas bo'ladi. Qisqarmas oddiy kasr shu kasrning qisqarmas shakli deb ham ataladi.

To'g'ri kasrning maxraji suratidan katta; noto'g'ri kasrning surati maxrajidan katta yoki teng. Noto'g'ri kasrlarni aralash son ko'rinishida ifodalash mumkin.

1-MISOL · QISQARMAS SHAKL

$\frac{14}{15}$: EKUB(14; 15) = 1 — qisqarmas.
$-\frac{12}{18}$: EKUB(12; 18) = 6, $-\frac{12}{18} = -\frac{2}{3}$.

2-MISOL · ARALASH SON

$-\frac{17}{5}$ — noto'g'ri kasr: $17 : 5 = 3$ (qoldiq 2).
$-\frac{17}{5} = -3\frac{2}{5}$.

QOIDA · QO'SHISH VA AYIRISH

Butun sonlarni qo'shish va ayirish qoidalari oddiy kasrlar uchun ham amal qiladi.

3-MISOL · HAR XIL ISHORALI KASRLAR

$-\frac{3}{4} + \frac{1}{6}$. EKUK(4; 6) = 12:
$-\frac{9}{12} + \frac{2}{12} = \frac{-9+2}{12} = -\frac{7}{12}$.

4-MISOL · MANFIY ARALASH SONNI AYIRISH

$1\frac{2}{3} - \left(-2\frac{1}{2}\right) = \frac{5}{3} + \frac{5}{2}$
$= \frac{10}{6} + \frac{15}{6} = \frac{25}{6} = 4\frac{1}{6}$.

QOIDA · KO'PAYTIRISH VA BO'LISH

Bir oddiy kasrni ikkinchi oddiy kasrga bo'lish birinchi kasrni ikkinchi kasrning teskarisiga ko'paytirishga teng kuchli.

Oddiy kasrlarni ko'paytirish yoki bo'lish uchun ularning maxrajlari bir xil songa keltirilishi shart emas.

5-MISOL · KO'PAYTIRISH

$-1\frac{1}{3} \cdot \frac{9}{10} = -\frac{4}{3} \cdot \frac{9}{10}$. Avval qisqartiramiz: 4 va 10 ni 2 ga, 9 va 3 ni 3 ga:

$$-\frac{2 \cdot 3}{1 \cdot 5} = -\frac{6}{5} = -1\frac{1}{5}.$$

6-MISOL · BO'LISH

$$-2\frac{1}{4} : \left(-1\frac{1}{2}\right) = -\frac{9}{4} : \left(-\frac{3}{2}\right)$$
$$= -\frac{9}{4} \cdot \left(-\frac{2}{3}\right) = \frac{3}{2} = 1\frac{1}{2}$$

(ishoralar bir xil — musbat).

QOIDA · O'NLI KASRLARNI QO'SHISH VA AYIRISH

Butun sonlarni qo'shish va ayirish qoidalari ratsional sonlar uchun ham taalluqlidir.

Usul. Ustun usulida vergullar bir ustunga yoziladi. Verguldan keyingi raqamlar soni teng bo'lishi uchun oxiriga nollar qo'yiladi (rasmda — to'q sariq). Raqamlar o'ngdan chapga qo'shiladi (ayiriladi).

7-MISOL · $8{,}46 + (-12{,}3)$

Ishoralar har xil: katta moduldan kichigini ayiramiz, ishora — moduli kattaniki.

$$\begin{array}{r} 12{,}30 \\ -\ 8{,}46 \\ \hline 3{,}84 \end{array}$$

$8{,}46 + (-12{,}3) = -3{,}84$.

8-MISOL · $-5{,}7 - 2{,}385$

$-5{,}7 - 2{,}385 = -5{,}7 + (-2{,}385)$ — ikkala son manfiy: modullarni qo'shamiz.

$$\begin{array}{r} 5{,}700 \\ +\ 2{,}385 \\ \hline 8{,}085 \end{array}$$

$-5{,}7 - 2{,}385 = -8{,}085$.

USUL · O'NLI KASRLARNI KO'PAYTIRISH

1-usul. Vergullarga qaramay, sonlarni butun sonlardek ko'paytiramiz. Natijada verguldan keyin shuncha raqam ajratamiz, ko'paytuvchilarda verguldan keyin jami nechta raqam bo'lsa.

2-usul. O'nli kasrlarni oddiy kasrga aylantirib ko'paytiramiz.

Ishora butun sonlardagidek: musbat · manfiy = manfiy, manfiy · manfiy = musbat.

9-MISOL · $2{,}6 \cdot (-1{,}3)$

$$\begin{array}{rl} 2{,}6 & \text{1 xona} \\ \times\ 1{,}3 & \text{1 xona} \\ \hline 7\,8 & \\ +\ 2\,6 & \\ \hline 3{,}3\,8 & \text{1 + 1 = 2 xona} \end{array}$$

Ishoralar har xil: $2{,}6 \cdot (-1{,}3) = -3{,}38$.

10-MISOL · $(-0{,}04) \cdot (-0{,}7)$

2-usul: $\frac{4}{100} \cdot \frac{7}{10} = \frac{28}{1000} = 0{,}028$.

$$\begin{array}{rl} 0{,}0\,4 & \text{2 xona} \\ \times\ 0{,}7 & \text{1 xona} \\ \hline 0{,}0\,2\,8 & \text{2 + 1 = 3 xona} \end{array}$$

Ishoralar bir xil: $(-0{,}04) \cdot (-0{,}7) = 0{,}028$.

USUL · O'NLI KASRGA BO'LISH

Bo'luvchini butun songa aylantiramiz.

1-usul. Bo'luvchida vergul nechta xona o'ngga surilsa, bo'linuvchida ham vergul shuncha xona o'ngga suriladi.

2-usul. Bo'linmani kasr ko'rinishida yozib, surat va maxrajini 10, 100, 1000, ... ga ko'paytiramiz.

11-MISOL · $(-5{,}46) : 0{,}6$

1-usul: 0,6 da vergul 1 xona o'ngga: 6. Bo'linuvchida ham: 54,6.
$54{,}6 : 6 = 9{,}1$. Manfiy : musbat = manfiy: $-9{,}1$.

12-MISOL · $(-0{,}315) : (-0{,}07)$

2-usul: $\frac{0{,}315}{0{,}07} = \frac{0{,}315 \cdot 100}{0{,}07 \cdot 100} = \frac{31{,}5}{7} = 4{,}5$.
Ishoralar bir xil — musbat: $4{,}5$.

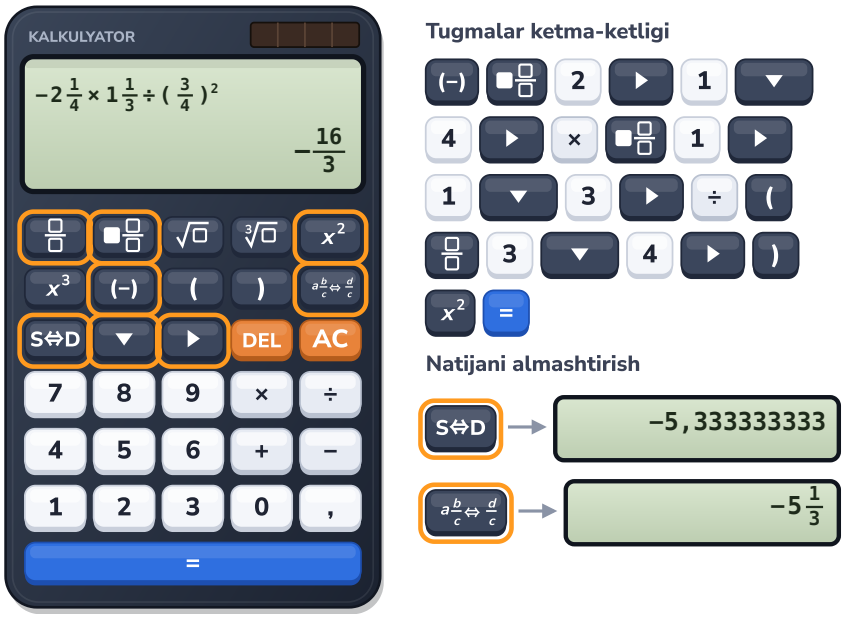
KALKULYATORDAN FOYDALANISH

Kalkulyatorda ifoda chapdan o'ngga, yozilishidagi tartibda teriladi. Manfiy son $(-)$ tugmasi bilan, kasr va aralash son — kasr tugmalari bilan kiritiladi.

S⇔D tugmasi oddiy va o'nli kasrlarni, $a\dfrac{b}{c} \Leftrightarrow \dfrac{d}{c}$ tugmasi esa aralash son va noto'g'ri kasrlarni bir-biriga o'tkazadi.

Kvadrat va kub — x^2, x^3 tugmalari, ildizlar — $\sqrt{}$ va $\sqrt[3]{}$ tugmalari, boshqa daraja — $x^{\square}$ tugmasi bilan: 4^{80} uchun 4, $x^{\square}$, 80, = bosiladi.

13-MISOL · $-2\dfrac{1}{4} \cdot 1\dfrac{1}{3} : \left(\dfrac{3}{4}\right)^2$



Qo'lda tekshiramiz: $-\dfrac{9}{4} \cdot \dfrac{4}{3} = -3$, $-3 : \dfrac{9}{16} = -3 \cdot \dfrac{16}{9} = -\dfrac{16}{3} = -5\dfrac{1}{3}$.

1-QISMDAGI SAVOLLAR

1-savol. $-18{,}4 + 5{,}7 = -(18{,}4 - 5{,}7) = -12{,}7$ °C.

2-savol. S⇔D tugmasini bosadi: $-\dfrac{45}{8} = -5{,}625$. $a\dfrac{b}{c} \Leftrightarrow \dfrac{d}{c}$ tugmasi esa $-5\dfrac{5}{8}$ ni ko'rsatadi.

4-QISM

Qo'llaymiz

1-MISOL · KASRLAR VA AMALLAR TARTIBI

Kalkulyatordan foydalanmasdan hisoblang: $-\frac{5}{6} : \left(\frac{1}{4} - \frac{2}{3}\right)$.

1. Avval qavs: $\frac{1}{4} - \frac{2}{3} = \frac{3}{12} - \frac{8}{12} = -\frac{5}{12}$.
2. Bo'lishni teskari songa ko'paytirish bilan almashtiramiz: $-\frac{5}{6} \cdot \left(-\frac{12}{5}\right)$.
3. Ishoralar bir xil — musbat. 5 va 5 ni, 6 va 12 ni qisqartiramiz: $\frac{1 \cdot 2}{1 \cdot 1} = 2$.

2-MISOL · ULUSHLAR

Bog'dagi daraxtlarning $\frac{2}{5}$ qismi olma, $\frac{1}{3}$ qismi o'rik, qolgani gilos. O'rik daraxtlarining $\frac{3}{8}$ qismi yosh ko'chatlar.

a) Gilos daraxtlari bog'ning qancha qismi? b) Yosh o'rik ko'chatlari bog'ning qancha qismi? c) Gilos 28 tup bo'lsa, bog'da nechta daraxt bor?



a. Butun bog' — 1. EKUK(5; 3) = 15: $1 - \frac{2}{5} - \frac{1}{3} = \frac{15 - 6 - 5}{15} = \frac{4}{15}$.

b. O'riklarning $\frac{3}{8}$ qismi — ko'paytiramiz: $\frac{1}{3} \cdot \frac{3}{8} = \frac{1}{8}$.

c. $\frac{4}{15}$ qismi — 28 tup, demak $\frac{1}{15}$ qismi $28 : 4 = 7$ tup. Butun bog': $7 \cdot 15 = 105$ tup.

3-MISOL · O'NLI KASRLAR VA AMALLAR TARTIBI

$(1{,}2 - 3{,}45) \cdot 0{,}4 : (-0{,}09)$

1. Qavs: $1{,}2 - 3{,}45 = -(3{,}45 - 1{,}20) = -2{,}25$.
2. Chapdan o'ngga: $-2{,}25 \cdot 0{,}4 = -0{,}9$ (xonalar: 2 + 1 = 3, $225 \cdot 4 = 900$ → 0,900).
3. $-0{,}9 : (-0{,}09) = 90 : 9 = 10$ (vergul ikkala sonda 2 xona o'ngga).

4-MISOL · MASALA

Tajribada muz bo'lagining harorati −7,2 °C. U har minutda 0,45 °C ga isiydi. Necha minutda harorati 0 °C ga yetadi?

1. Harorat qanchaga ko'tarilishi kerak: $0 - (-7{,}2) = 7{,}2$ °C.
2. $7{,}2 : 0{,}45 = 720 : 45 = 16$ minut.

5-MISOL · KALKULYATOR BILAN

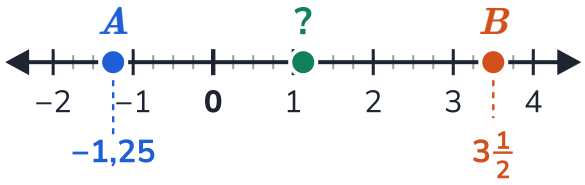
$\sqrt[3]{-3{,}375} + 1{,}5^2 \cdot 0{,}8$



✓ Qo'lda: $(-1{,}5)^3 = -3{,}375$, demak $\sqrt[3]{-3{,}375} = -1{,}5$; $2{,}25 \cdot 0{,}8 = 1{,}8$; $-1{,}5 + 1{,}8 = 0{,}3$.

6-MISOL · IKKI SONDAN TENG UZOQLIKDAGI SON

$-1{,}25$ va $3\frac{1}{2}$ dan bir xil uzoqlikda joylashgan sonni toping. Javobni oddiy kasr ko'rinishida yozing.



1. Bu son ikki sonning o'rtasida — ularning yig'indisining yarmi.
2. $-1{,}25 = -\frac{5}{4}, 3\frac{1}{2} = \frac{14}{4}: \left(-\frac{5}{4} + \frac{14}{4}\right) : 2 = \frac{9}{4} \cdot \frac{1}{2} = \frac{9}{8}.$
3. $\frac{9}{8} = 1{,}125$: $1{,}125 - (-1{,}25) = 2{,}375$ va $3{,}5 - 1{,}125 = 2{,}375$.



Ko'paytirishda vergullarni tekislash shart emas — xonalarni sanash kerak!

DIQQAT!

1. ✗ $0{,}3 \cdot 0{,}3 = 0{,}9$ ✓ $0{,}3 \cdot 0{,}3 = 0{,}09$ **(1 + 1 = 2 xona)**
2. ✗ $-2\frac{1}{2} \cdot 4 = -8\frac{1}{2}$ ✓ $-\frac{5}{2} \cdot 4 = -10$ **— avval noto'g'ri kasrga aylantiramiz**
3. ✗ $\frac{2}{3} : \frac{4}{5} = \frac{2}{3} \cdot \frac{4}{5}$ ✓ $\frac{2}{3} \cdot \frac{5}{4} = \frac{5}{6}$ **— ikkinchi kasr teskarisiga almashadi**
4. ✗ $1{,}5 : 0{,}05 = 3$ ✓ $150 : 5 = 30$ **— vergul ikkala sonda 2 xona suriladi**
5. Kalkulyator juda katta sonlarni ko'rsata olmaydi: 5^{140} ni hisoblaydi, 5^{150} da esa xato (ERROR) chiqadi.

Mashqlar

1. Qisqarmas kasr ko'rinishida yozing; noto'g'ri kasrni aralash son qiling.
   a) $-\frac{15}{35}$ b) $-\frac{36}{48}$ c) $\frac{22}{9}$ d) $\frac{21}{40}$ e) $-\frac{50}{15}$

2. Kalkulyatordan foydalanmasdan hisoblang.
   a) $\frac{2}{9}+\frac{5}{12}$ b) $-\frac{3}{5}+\left(-\frac{1}{4}\right)$ c) $3\frac{1}{6}-4\frac{3}{8}$ d) $-\frac{5}{6}-(-2)+\left(-\frac{3}{4}\right)$

3. Kalkulyatordan foydalanmasdan hisoblang.
   a) $-\frac{14}{27}\cdot\frac{9}{35}$ b) $\frac{5}{12}\cdot 2\frac{2}{5}\cdot 1\frac{1}{4}$ c) $-3\frac{3}{4}:\left(-1\frac{7}{8}\right)$ d) $\frac{4}{15}:\left(-2\frac{2}{5}\right)$
   e) $\frac{1}{5}-\left(-\frac{3}{5}\right)^2$ f) $\frac{-1\frac{1}{2}}{2\frac{1}{4}}$ g) $1\frac{1}{3}:\left(-\frac{2}{9}\right)-2$

4. Kalkulyatordan foydalanmasdan hisoblang.
   a) $18{,}46+7{,}9$ b) $31{,}05-12{,}768$ c) $-4{,}6-7{,}25$ d) $6{,}03-(-9{,}097)$ e) $20+(-3{,}64)$

5. Kalkulyatordan foydalanmasdan hisoblang.
   a) $2{,}4\cdot 3{,}6$ b) $0{,}125\cdot(-0{,}08)$ c) $(-7{,}5)\cdot(-0{,}44)$
   d) $(-4{,}32):0{,}9$ e) $(-0{,}2736):(-0{,}006)$ f) $17{,}1:0{,}45$

6. Hisoblang. Javobni qisqarmas kasr yoki butun son ko'rinishida yozing.
   a) $\left(-\frac{2}{3}\right)^2:\left(\frac{1}{3}-\frac{5}{6}+\frac{1}{4}\right)$ b) $2\frac{1}{4}-\frac{1}{2}\cdot\left(-1\frac{1}{2}\right):\frac{5}{6}$
   c) $-1\frac{2}{5}\cdot\left(-\frac{5}{14}+\frac{3}{4}\right)$ d) $\left(-\frac{1}{2}\right)^3-\left[-2\frac{1}{3}-\left(-\frac{1}{2}\right)\right]:4$

7. Kalkulyatordan foydalanmasdan hisoblang.
   a) $2{,}94:1{,}4-5{,}2$ b) $8:(0{,}6+1{,}9)\cdot 3{,}2$
   c) $(2{,}3-6{,}1):1{,}9\cdot(-0{,}7)$ d) $2{,}5\cdot[-4{,}36+2{,}1:(-0{,}7)]$

8. Kalkulyatordan foydalanib hisoblang.
   a) $\sqrt{13^2+84^2}$ b) $\sqrt[3]{-2744}:\left(-1\frac{3}{4}\right)^2$ c) $\left(1\frac{1}{2}-2\frac{3}{4}\right)^3\cdot 4$
   d) $\frac{1{,}2^2-2{,}9}{0{,}4\cdot(-0{,}5)}$ e) $1{,}75\cdot 6{,}4-(-1{,}2)^2\cdot 5$

9. Kalkulyatorda hisoblang va ekrandagi javobning dastlabki 5 ta raqamini yozing.
   a) $4{,}13 \cdot \left(-1\frac{2}{3}\right)^2$ b) $\left(-\frac{7}{3}\right)^3 : \left(\frac{2}{5}\right)^2$ c) $\sqrt{7{,}3} - (-1{,}2)^3$

10. Kalkulyator 10^{100} dan kichik sonlarnigina ko'rsata oladi. 4^{10}, 4^{80}, 4^{200} dan qaysi birini hisoblab bo'lmaydi? Kalkulyatorda tekshiring. 4^{80} ning javobidagi «×10^{48}» nimani bildiradi?

11. Ekin maydonining $\frac{3}{10}$ qismiga bug'doy, qolgan qismining $\frac{4}{7}$ qismiga paxta ekildi. Qolgan 45 ga yerga sabzavot ekildi.
    a) Paxta maydonning qancha qismini egallaydi? b) Ekin maydoni necha gektar?

12. Bog'bon har bir ko'chatga $\frac{3}{8}$ l dori eritmasi sepadi. Purkagichda 2,5 l eritma bor.
    a) 6 ta ko'chatga eritma yetadimi? b) Ko'pi bilan nechta ko'chatga sepish mumkin? Qancha eritma ortib qoladi?

13. Usta simni uch bo'lakka kesdi: birinchi bo'lak butun simning $\frac{1}{4}$ qismi, ikkinchisi $\frac{2}{5}$ qismi, uchinchisi — qolgani. Ikkinchi bo'lak birinchisidan 2,7 m uzun chiqdi.
    a) Qaysi bo'lak eng uzun? b) Sim necha metr bo'lgan? Uchinchi bo'lak necha metr?

14. Yuk lifti ko'pi bilan 0,6 t ko'taradi. Liftga 82,5 kg li yukchi kirdi. U har biri 0,035 t bo'lgan qutilarni chiqarmoqchi. Bir safarda ko'pi bilan nechta quti ola oladi?

15. Son o'qida $A(-2{,}375)$ va $B\left(1\frac{1}{8}\right)$ nuqtalar belgilangan. AB kesmaning o'rtasi qaysi son? Oddiy kasr ko'rinishida yozing.

16. Mashinaning benzin ko'rsatkichi $\frac{1}{4}$ va $\frac{1}{2}$ belgilarining aynan o'rtasida turibdi (0 — bo'sh, 1 — to'la). Bakka 60 l benzin sig'adi. Mashina 1 l benzin bilan 12 km yuradi.
    a) Bakning qancha qismida benzin bor? b) Shu benzin bilan mashina necha kilometr yura oladi?

17. Bir chekli o'nli kasrni ikkinchisiga bo'ldik. Bo'linma qanday son chiqishi mumkin: a) butun son; b) chekli o'nli kasr; c) davriy o'nli kasr; d) irratsional son? Har biriga misol keltiring yoki nega bo'lmasligini ayting.

18. Do'konda 1 kg guruch 14 500 so'm, 1 l yog' 21 000 so'm. Onaning bank kartasida −3 200 so'm (qarz) bor edi, kartaga 80 000 so'm tushdi.
    a) Kartada endi necha so'm bor? b) Ona 3 kg guruch va 1 l yog' oldi. Kartada qancha pul qoldi? c) Shu ma'lumotlar bo'yicha javobi 3 bo'ladigan savol tuzing.

5-QISM

Tekshiramiz

1. Qisqarmas shaklda yozing: a) $-\frac{24}{40}$ b) $\frac{35}{12}$ (aralash son ko'rinishida).
2. Hisoblang: a) $-\frac{5}{8}+\frac{1}{6}$ b) $1\frac{3}{4}-\left(-2\frac{2}{3}\right)$ c) $-2\frac{2}{5}\cdot\frac{5}{6}$ d) $-\frac{3}{10}:\left(-1\frac{4}{5}\right)$
3. Hisoblang: a) $9{,}7+(-12{,}45)$ b) $-3{,}08-0{,}9$ c) $1{,}5\cdot(-0{,}26)$ d) $(-0{,}684):0{,}04$
4. Hisoblang: $\left(-\frac{2}{3}+\frac{1}{4}\right):\left(-\frac{5}{6}\right)^2$.
5. Hisoblang: $(3{,}6-5{,}1)\cdot 0{,}8:(-0{,}3)$.
6. a) Kalkulyatorda $\sqrt[3]{-2{,}197}+2{,}4^2$ ni hisoblang.

   b) Ekranda 2,375 turibdi. S⇔D tugmasi bosilsa, nima chiqadi? Keyin $a\frac{b}{c}\Leftrightarrow\frac{d}{c}$ bosilsa-chi?
7. Sinf o'quvchilarining $\frac{1}{3}$ qismi shaxmat, $\frac{2}{9}$ qismi kurash to'garagiga, qolganlari futbolga qatnashadi. Shaxmatchilarning $\frac{3}{4}$ qismi o'g'il bolalar.

   a) Futbolchilar sinfning qancha qismi? b) Shaxmat o'ynaydigan o'g'il bolalar sinfning qancha qismi?
8. Qaysi son −1,6 ga ham, $2\frac{1}{5}$ ga ham bir xil masofada joylashgan?
9. **Xatoni toping.**

   a) Sardor: «$0{,}2\cdot 0{,}4=0{,}8$». b) Dilnoza: «$-1\frac{1}{3}\cdot 3=-3\frac{1}{3}$». c) Bekzod: «$2{,}4:0{,}08=3$».

- [ ] Men manfiy kasr va aralash sonlarni qo'sha, ayira, ko'paytira va bo'la olaman.
- [ ] Men manfiy o'nli kasrlar bilan to'rt amalni bajara olaman (vergul, xonalar, ishora).
- [ ] Men ifodani kalkulyatorda hisoblab, javobni oddiy kasr, o'nli kasr va aralash son ko'rinishiga o'tkaza olaman.
- [ ] Men ulushga oid masalalarni yecha olaman.

Javoblar

4-qism · Mashqlar

1-masala

a) $-\frac{3}{7}$ b) $-\frac{3}{4}$ c) $2\frac{4}{9}$ d) qisqarmas: EKUB(21; 40) = 1 e) $-\frac{10}{3} = -3\frac{1}{3}$.

2-masala

a) $\frac{8}{36} + \frac{15}{36} = \frac{23}{36}$. b) $-\frac{12}{20} - \frac{5}{20} = -\frac{17}{20}$. c) $\frac{76}{24} - \frac{105}{24} = -\frac{29}{24} = -1\frac{5}{24}$. d) $-\frac{10}{12} + \frac{24}{12} - \frac{9}{12} = \frac{5}{12}$.

3-masala

a) $-\frac{2}{15}$. b) $\frac{5}{12} \cdot \frac{12}{5} \cdot \frac{5}{4} = \frac{5}{4} = 1\frac{1}{4}$. c) $\frac{15}{4} \cdot \frac{8}{15} = 2$. d) $\frac{4}{15} \cdot \left(-\frac{5}{12}\right) = -\frac{1}{9}$. e) $\frac{5}{25} - \frac{9}{25} = -\frac{4}{25}$. f) $-\frac{3}{2} \cdot \frac{4}{9} = -\frac{2}{3}$. g) $\frac{4}{3} : \left(-\frac{2}{9}\right) = -6$, $-6 - 2 = -8$.

4-masala

a) 26,36 b) 18,282 c) −11,85 d) 15,127 e) 16,36

5-masala

a) 8,64 b) −0,01 c) 3,3 d) $-43{,}2 : 9 = -4{,}8$ e) $273{,}6 : 6 = 45{,}6$ f) $1710 : 45 = 38$

6-masala

a) Qavs: $\frac{4}{12} - \frac{10}{12} + \frac{3}{12} = -\frac{1}{4}$; $\frac{4}{9} : \left(-\frac{1}{4}\right) = -\frac{16}{9} = -1\frac{7}{9}$. b) $\frac{1}{2} \cdot \left(-\frac{3}{2}\right) = -\frac{3}{4}$; $-\frac{3}{4} \cdot \frac{6}{5} = -\frac{9}{10}$; $\frac{9}{4} - \left(-\frac{9}{10}\right) = \frac{45}{20} + \frac{18}{20} = \frac{63}{20} = 3\frac{3}{20}$. c) Qavs: $-\frac{10}{28} + \frac{21}{28} = \frac{11}{28}$; $-\frac{7}{5} \cdot \frac{11}{28} = -\frac{11}{20}$. d) $\left(-\frac{1}{2}\right)^3 = -\frac{1}{8}$; $-2\frac{1}{3} + \frac{1}{2} = -\frac{11}{6}$, $-\frac{11}{6} : 4 = -\frac{11}{24}$; $-\frac{3}{24} + \frac{11}{24} = \frac{8}{24} = \frac{1}{3}$.

7-masala

a) $2{,}94 : 1{,}4 = 2{,}1$; $2{,}1 - 5{,}2 = -3{,}1$. b) $0{,}6 + 1{,}9 = 2{,}5$; $8 : 2{,}5 = 3{,}2$; $3{,}2 \cdot 3{,}2 = 10{,}24$. c) $2{,}3 - 6{,}1 = -3{,}8$; $-3{,}8 : 1{,}9 = -2$; $-2 \cdot (-0{,}7) = 1{,}4$. d) $2{,}1 : (-0{,}7) = -3$; $-4{,}36 + (-3) = -7{,}36$; $2{,}5 \cdot (-7{,}36) = -18{,}4$.

8-masala

a) $\sqrt{169 + 7056} = \sqrt{7225} = 85$. b) $-14 : \dfrac{49}{16} = -14 \cdot \dfrac{16}{49} = -\dfrac{32}{7} = -4\dfrac{4}{7}$. c) $\left(-1\dfrac{1}{4}\right)^3 \cdot 4 = -\dfrac{125}{64} \cdot 4 = -\dfrac{125}{16} = -7\dfrac{13}{16}$. d) $\dfrac{1{,}44 - 2{,}9}{-0{,}2} = \dfrac{-1{,}46}{-0{,}2} = 7{,}3$. e) $11{,}2 - 1{,}44 \cdot 5 = 11{,}2 - 7{,}2 = 4$.

9-masala

a) 11,472 ($4{,}13 \cdot \dfrac{25}{9} = 11{,}4722\ldots$). b) −79,398 ($-\dfrac{343}{27} : \dfrac{4}{25} = -\dfrac{8575}{108} = -79{,}398\ldots$). c) 4,4298 ($2{,}70185\cdots + 1{,}728$).

10-masala

$4^{10} = 1\,048\,576$. 4^{80} juda katta: ekranda «1,461501637 ×10^{48}» chiqadi — bu 1,461501637 ni 10 ning 48-darajasiga ko'paytirish degani (javob taqribiy). 4^{200} da xato (ERROR) chiqadi: $4^{200} \approx 2{,}6 \cdot 10^{120}$ — kalkulyator ko'rsata oladigan eng katta sondan katta.

11-masala

a) Qolgan qism $\dfrac{7}{10}$; paxta: $\dfrac{7}{10} \cdot \dfrac{4}{7} = \dfrac{4}{10} = \dfrac{2}{5}$. b) Sabzavot: $1 - \dfrac{3}{10} - \dfrac{4}{10} = \dfrac{3}{10}$. $\dfrac{3}{10}$ qismi 45 ga, demak $\dfrac{1}{10}$ qismi 15 ga, butun maydon $15 \cdot 10 = 150$ ga.

12-masala

a) $6 \cdot \dfrac{3}{8} = \dfrac{18}{8} = 2{,}25$ l — 2,5 l dan kam, yetadi. b) $2{,}5 : \dfrac{3}{8} = \dfrac{5}{2} \cdot \dfrac{8}{3} = \dfrac{20}{3} = 6\dfrac{2}{3}$ — butun ko'chatlar soni 6 ta. Ortadi: $2{,}5 - 2{,}25 = 0{,}25$ l.

13-masala

a) Uchinchi bo'lak: $1 - \dfrac{1}{4} - \dfrac{2}{5} = \dfrac{20 - 5 - 8}{20} = \dfrac{7}{20}$. Bo'laklar: $\dfrac{5}{20}$, $\dfrac{8}{20}$, $\dfrac{7}{20}$ — eng uzuni ikkinchi bo'lak. b) Farq: $\dfrac{2}{5} - \dfrac{1}{4} = \dfrac{3}{20}$ — bu 2,7 m. $\dfrac{1}{20}$ qismi $2{,}7 : 3 = 0{,}9$ m, sim $0{,}9 \cdot 20 = 18$ m. Uchinchi bo'lak: $0{,}9 \cdot 7 = 6{,}3$ m.

14-masala

0,6 t = 600 kg, 0,035 t = 35 kg. Qutilarga: $600 - 82{,}5 = 517{,}5$ kg. $517{,}5 : 35 = 14{,}78\ldots$ — ko'pi bilan **14 ta** quti (15 tasi og'irlik qiladi: $15 \cdot 35 = 525$ kg).

15-masala

$-2{,}375 = -\frac{19}{8}$; $\left(-\frac{19}{8} + \frac{9}{8}\right) : 2 = -\frac{10}{8} \cdot \frac{1}{2} = -\frac{5}{8}$.

16-masala

a) $\left(\frac{1}{4} + \frac{1}{2}\right) : 2 = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$. b) Benzin: $60 \cdot \frac{3}{8} = 22{,}5$ l; yo'l: $22{,}5 \cdot 12 = 270$ km.

17-masala

a) Butun son: $0{,}6 : 0{,}2 = 3$. b) Chekli o'nli kasr: $0{,}6 : 0{,}4 = 1{,}5$. c) Davriy o'nli kasr: $0{,}1 : 0{,}3 = \frac{1}{3} = 0{,}(3)$. d) Irratsional son chiqmaydi: bo'linma har doim ikki butun sonning nisbati, ya'ni ratsional son.

18-masala

a) $-3\,200 + 80\,000 = 76\,800$ so'm. b) Xarid: $3 \cdot 14\,500 + 21\,000 = 64\,500$ so'm; qoldi: $76\,800 - 64\,500 = 12\,300$ so'm. c) Masalan: «a) dagi pulga ko'pi bilan necha litr yog' olish mumkin?» — $76\,800 : 21\,000 = 3{,}65\ldots$, ya'ni 3 l.

5-qism · Tekshiramiz

1-masala

a) $-\frac{3}{5}$ b) $2\frac{11}{12}$

2-masala

a) $-\frac{15}{24} + \frac{4}{24} = -\frac{11}{24}$. b) $\frac{21}{12} + \frac{32}{12} = \frac{53}{12} = 4\frac{5}{12}$. c) $-\frac{12}{5} \cdot \frac{5}{6} = -2$. d) $-\frac{3}{10} \cdot \left(-\frac{5}{9}\right) = \frac{1}{6}$.

3-masala

a) −2,75 b) −3,98 c) −0,39 d) $-68{,}4 : 4 = -17{,}1$

4-masala

$-\frac{8}{12} + \frac{3}{12} = -\frac{5}{12}$; $\left(-\frac{5}{6}\right)^2 = \frac{25}{36}$; $-\frac{5}{12} \cdot \frac{36}{25} = -\frac{3}{5}$.

5-masala

$-1{,}5 \cdot 0{,}8 = -1{,}2$; $-1{,}2 : (-0{,}3) = 4$.

6-masala

a) $-1{,}3 + 5{,}76 = 4{,}46$. b) S⇔D: $\frac{19}{8}$; $a\frac{b}{c} \Leftrightarrow \frac{d}{c}$: $2\frac{3}{8}$.

7-masala

a) $1 - \frac{3}{9} - \frac{2}{9} = \frac{4}{9}$. b) $\frac{1}{3} \cdot \frac{3}{4} = \frac{1}{4}$.

8-masala

$(-1{,}6 + 2{,}2) : 2 = 0{,}3$.

9-masala

a) 1 + 1 = 2 xona: $0{,}2 \cdot 0{,}4 = 0{,}08$. b) $-\frac{4}{3} \cdot 3 = -4$. c) Vergul ikkala sonda 2 xona: $240 : 8 = 30$.